

SBC 201 Chapter 35 Referenced Standards — Group R-2 High-Rise Dimensional Requirements Matrix (2024)

1. Document metadata and use limitation

- **Project basis:** Riyadh, Saudi Arabia; Group R-2 residential high-rise.
- **Deliverable tier:** Project-use matrices in Sections 1–9 (design-check rows, not pasted inventory), plus a coverage summary and unresolved-source register. The full row inventory is not published.
- **Code/source basis:** SBC 201 (2024), Chapter 35, source file Reference\SBC 201 2024\source_reference\Chapter_35 – REFERENCED STANDARDS.txt.
- **Extraction audit:** Skill-finetune run. Project-use rows follow the chapter-extract row contract (noun-phrase checks, bold published tokens, building-language triggers, named exceptions, check-specific actions). Internal inventory: **0** independently checkable numeric records (**0** Verified, **0** Verify source). Edition years, standard numbers and citing-section numbers are identifiers, not dimensional design limits. Title-embedded quantities are outbound and are not adopted. Agency tables were read; General Comments and Purpose are commentary and were excluded.
- **Model:** Cursor Grok 4.6.
- **Prepared:** 2026-08-27.
- **Status:** Source-only architectural advisory matrix for design coordination. It is not a stamped compliance statement, specification, SCD NOC, SBPS approval, legal opinion, or permission to construct.
- **Outbound-source rule:** No value in this matrix has been imported from NFPA, ASME, ICC A117.1, ASTM test methods, UL listings, SBC 304/305, Chapter 9/10/11/16, commentary examples, or the existing chapter summary. Where Chapter 35 names a companion, this matrix records the listed designation and year without supplying that companion’s criteria.

Scope and assumptions

1. Group R-2 and high-rise status are project statements, not independently verified classifications.
2. The exact Riyadh AHJ/permit pathway, project stage and SCD NOC status are unconfirmed; therefore this matrix does not conclude compliance.
3. Automatic sprinkler protection (**NFPA 13—13** versus **NFPA 13R—13**), storey count, mixed-use podium, amenity program, structural system and façade combustibility are unconfirmed. Both listed sprinkler companions are shown until the project locks one. **NFPA 13D—13** is omitted as a one- and two-family path.
4. Chapter 35 is an edition list. It does not publish millimetre geometry, sprinkler densities, fire-resistance hours, elevator dimensions or accessibility clearances.
5. Listed standards are for guidance until adopted and/or approved by **SASO** or **SBCNC**. Application is through **Section 102.4**, whose text is not in this chapter.

2. Legends

Applicability

Status	Meaning
Direct	Expected to govern the stated R-2 tower basis, subject to confirmed geometry and design data.
Conditional	Governs only when the stated feature, use, occupant load, sprinkler branch or exception exists.
Not typical	Unrelated occupancy-only rule; omitted from this deliverable unless the gap register already opened that use.
External verification	Chapter 35 points to another section/code/standard, or the project/AHJ basis must be confirmed before use.

Source confidence

Status	Meaning
Verified	Requirement and any stated numeric value were checked against unambiguous mandatory Chapter 35 source text or an unambiguous table cell.
Verify source	OCR, flattened table, page-split, or footnote attachment is unresolved. Not a design-release value.

3. Project decision and gap register

Decision / gap	Current project basis	Why it controls Chapter 35 application	Required project action
SASO / SBCNC adoption	Unconfirmed for each listed edition	Listed standards are guidance until adopted and/or approved by SASO or SBCNC	AHJ/code consultant to confirm which Chapter 35 editions are adopted for enforcement
Section 102.4 application	Text not in this chapter	Chapter 35 sends application of every listed standard to Section 102.4	Apply companions through Section 102.4; do not reconstruct 102.4 from this matrix
NFPA 13 versus 13R	Unconfirmed	NFPA 13—13 and NFPA 13R—13 are both listed; this chapter does not assign either path to Group R-2 high-rise	Lock the sprinkler standard from Chapter 9 / the fire strategy; keep both editions on the branch list until then
Mixed-use / podium / amenity	Unconfirmed	Non-residential portions can add cooking-suppression, assembly, parking-lift or other listed companions omitted here as not typical	Freeze occupancy by space before adding those designations to the spec sheet
Storey count / high-rise systems	Unconfirmed	Standpipe, fire pump, smoke control, emergency power and fire-service access elevator companions apply only if those systems exist	Confirm the systems on the life-safety drawings, then name the listed editions
Accessibility and elevator editions	Listed here as ICC A117.1—09 and ASME/A17.1—13 / CSA B44—2013	Spec sheets must use the printed years, not a later edition from memory	Put the Chapter 35 designation-and-year on the accessibility and elevator specifications
Structural system	Unconfirmed (concrete typical; steel, masonry, aluminum, cold-formed optional)	SBC 304 , SBC 305 , AISC 341—10 , ADM1—2015 and AISI S100—12 are separate listings	Structural engineer to name only the companions that match the locked frame
Façade combustibility / roof type	Unconfirmed	NFPA 285—12 , NFPA 268—12 , FM/UL/SPRI roof tests apply only to the matching assembly	Façade and roof consultants to select the listed test editions for the actual build-up
NOC / fire strategy	Unconfirmed	Companion fire standards cannot evidence SCD NOC from this chapter	Engage the fire/MEP engineer of record before freezing the specification editions

4. List application

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
Chapter 35	Referenced-standards list	This chapter lists the standards referenced in this document by promulgating agency, standard identification, effective date, title, and citing section(s)	All SBC 201 references to listed standards	Listed for guidance until the SASO/SBCNC adoption gate	Direct	Put Chapter 35 on the spec sheet as the edition list for every named companion	Verified
Chapter 35	Referenced-standard application	Application of the referenced standards shall be as specified in Section 102.4	Every use of a Chapter 35 listed standard	None stated	External verification	Apply each listed standard through Section 102.4 ; do not import 102.4 text or companion criteria into this matrix	Verified
Chapter 35	SASO/SBCNC enforcement gate	Standards are listed for guidance; for enforcement they shall be adopted and/or approved by SASO or SBCNC as per their mandates before being considered part of this code	Enforcement of any listed standard	None stated	External verification	Confirm SASO or SBCNC adoption of each specified edition before treating it as enforceable	Verified

5. Life-safety companions

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
NFPA 13—13	NFPA 13 sprinkler edition	Installation of Sprinkler Systems as listed: NFPA 13—13	Automatic sprinkler system charged to this designation	Do not substitute NFPA 13R—13 or NFPA 13D—13 unless the charging section names that standard	Conditional	Name NFPA 13—13 on the sprinkler specification if Chapter 9 locks this path; do not import densities, spacing or area limits from NFPA 13	Verified
NFPA 13R—13	NFPA 13R sprinkler edition	Installation of Sprinkler Systems in Low Rise Residential Occupancies as listed: NFPA 13R—13	Low-rise residential sprinkler system charged to this designation	Not a substitute for NFPA 13—13 . NFPA 13D—13 is a one- and two-family path and is omitted here	Conditional	Keep NFPA 13R—13 on the branch list until Chapter 9 locks 13 versus 13R; do not import 13R criteria into this matrix	Verified
NFPA 14—13	Standpipe installation edition	Installation of Standpipe and Hose System as listed: NFPA 14—13	Standpipe and hose system	None stated	Conditional	Name NFPA 14—13 on the standpipe specification; do not import hose, outlet or pressure figures from NFPA 14	Verified
NFPA 20—13	Fire-pump installation edition	Installation of Stationary Pumps for Fire Protection as listed: NFPA 20—13	Stationary fire pump	None stated	Conditional	Name NFPA 20—13 on the fire-pump specification; do not import pump, driver or supply figures from NFPA 20	Verified
NFPA 72—13	Fire-alarm signaling edition	National Fire Alarm and Signaling Code as listed: NFPA 72—13	Fire alarm and signaling system	None stated	External verification	Name NFPA 72—13 on the fire-alarm specification; do not import detector spacing or candela figures from NFPA 72	Verified
NFPA 70—14	Electrical-code edition	National Electrical Code as listed: NFPA 70—14	Electrical design and installation charged to this designation	None stated	External verification	Name NFPA 70—14 on the electrical specification; do not import ampacity or wiring figures from NFPA 70	Verified
NFPA 10—13	Portable extinguisher edition	Portable Fire Extinguishers as listed: NFPA 10—13	Portable fire extinguishers	None stated	Conditional	Name NFPA 10—13 on the extinguisher specification; do not import rating or travel-distance figures from NFPA 10	Verified
NFPA 110—13; NFPA 111—13	Emergency-power editions	Emergency and standby power as listed: NFPA 110—13 ; stored electrical energy systems NFPA 111—13	Emergency or standby power system	None stated	Conditional	Name NFPA 110—13 and NFPA 111—13 on the emergency-power specification; do not import runtime or classification figures	Verified
NFPA 92—15	Smoke-control edition	Smoke Control Systems as listed: NFPA 92—15	Smoke-control system	None stated	Conditional	Name NFPA 92—15 if a smoke-control system is provided; do not import design-fire or pressure figures from NFPA 92	Verified
NFPA 720—15	CO-detection edition	Carbon monoxide detection and warning equipment as listed: NFPA 720—15	CO detection or warning equipment	Attached citing-section token is damaged (9151.6.1); designation and title remain readable	Conditional	Name NFPA 720—15 if CO detection is provided; confirm citing sections in Chapter 9; do not import CO siting figures	Verified

6. Egress, accessibility and elevators

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
ICC A117.1—09	Accessibility companion edition	Accessible and Usable Buildings and Facilities as listed: ICC A117.1—09	Accessibility technical criteria charged to this designation	Do not substitute a later A117.1 edition from memory	External verification	Name ICC A117.1—09 on the accessibility specification; do not import millimetre unit geometry from A117.1	Verified
ASME A17.1—13	Elevator safety-code edition	Safety Code for Elevators and Escalators as listed: ASME/A17.1—13 / CSA B44—2013	Elevators and escalators	Performance-based path A17.7—2007/CSA B44—07 is a separate listing	External verification	Name ASME A17.1—13 / CSA B44—2013 on the elevator specification; do not import cab, pit or door figures	Verified
ASME A18.1—2008	Platform-lift edition	Safety Standard for Platform Lifts and Stairway Chairlifts as listed: A18.1—2008	Platform lift or stairway chairlift	Not a substitute for ASME A17.1—13	Conditional	Name A18.1—2008 only if a platform lift or chairlift is provided; do not import lift geometry	Verified
BHMA A156.10—2011; A156.19—2013	Power-operated door editions	Power-operated pedestrian doors A 156.10—2011 and power-assist / low-energy doors A 156.19—2013 as listed	Power-operated or low-energy pedestrian door	Revolving doors are listed separately as A 156.27—11	Conditional	Name the matching BHMA A156 edition on the door specification; do not import opening-force or timing figures	Verified
ASTM F2090—10	Emergency-escape window guard	Window fall-prevention devices with emergency-escape release as listed: ASTM F2090—10	Windows that require fall-prevention devices and an emergency-escape release	Nonemergency-escape devices are listed as ASTM F2006 (edition token damaged in the attached table)	Conditional	Name ASTM F2090—10 on windows that must retain emergency escape; do not import opening limits from F2090	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
ASTM F2006	Nonemergency window guard	Window fall-prevention devices for nonemergency escape and rescue as listed; attached token reads F2006—00 (2005) 10	Windows that require fall-prevention devices without an emergency-escape release	Emergency-escape devices are ASTM F2090—10 . Do not adopt the damaged year token as a design-release edition	Conditional	Confirm the printed ASTM F2006 year against the code volume before specifying; do not import opening limits	Verify source
UL 294—1999; UL 305—2012	Access-control and panic editions	Access control system units UL 294—1999 (revisions through September 2010) and panic hardware UL 305—2012 as listed	Electrically locked egress doors or panic hardware	None stated	Conditional	Name UL 294 and UL 305 on the hardware specification as listed; do not import listing-test criteria	Verified
UL 924—06; UL 1994—04	Emergency lighting and path marking	Emergency lighting and power equipment UL 924—06 and luminous egress path marking UL 1994—04 as listed	Emergency lighting or luminous path marking	Photoluminescent markings also listed as ASTM E2072—10	Conditional	Name the listed UL/ASTM emergency-lighting and path-marking editions on the egress-lighting specification; do not import luminance figures	Verified
ANSI/ASSE Z359.1-2007	Personal fall-arrest edition	Personal fall arrest systems as listed: ANSI/ASSE Z359.1-2007	Fall-arrest systems charged to this designation	None stated	Conditional	Name ANSI/ASSE Z359.1-2007 on rooftop fall-arrest specifications; do not import anchorage figures	Verified

7. Fire-resistance and finish test methods

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
ASTM E119—2012A; UL 263—11	Construction fire-test editions	Fire tests of building construction and materials as listed: ASTM E119—2012A and UL 263—11	Fire-resistance-rated assemblies tested to these methods	None stated	External verification	Name ASTM E119—2012A and UL 263—11 on rated-assembly specifications; do not import hourly ratings from those test methods	Verified
ASTM E84—2013A; UL 723—2008	Surface-burning test editions	Surface burning characteristics as listed: ASTM E84—2013A and UL 723—2008	Interior finish and materials charged to these test methods	None stated	External verification	Name ASTM E84—2013A and UL 723—2008 on finish specifications; do not import flame-spread or smoke-developed figures	Verified
NFPA 80—13; NFPA 105—13	Fire and smoke door editions	Fire doors and opening protectives NFPA 80—13 and smoke door assemblies NFPA 105—13 as listed	Fire-rated or smoke door assemblies	Companion fire tests include NFPA 252—12 , NFPA 257—12 , UL 10B—2008 and UL 10C—2009	External verification	Name NFPA 80—13 and NFPA 105—13 on opening-protective specifications; do not import clearance or listing figures	Verified
UL 555—2006; UL 555C—2006; UL 555S—99	Damper test editions	Fire dampers UL 555—2006 , ceiling dampers UL 555C—2006 , and smoke dampers UL 555S—99 as listed	Duct and air-transfer opening protectives	None stated	Conditional	Name the matching UL 555-series listing on damper schedules; do not import leakage or hourly figures	Verified
NFPA 285—12; NFPA 268—12	Combustible exterior-wall tests	Exterior nonload-bearing wall assemblies containing combustible components NFPA 285—12 and exterior-wall ignitability NFPA 268—12 as listed	Combustible or foam-plastic exterior wall assemblies	Attached table concatenates NFPA 285—12 with NFPA 286—15 ; designations remain readable	Conditional	Name NFPA 285—12 and NFPA 268—12 if the façade contains combustible components; do not import fire-propagation criteria	Verified
NFPA 286—15; NFPA 253—15	Interior-finish and floor-covering tests	Room fire-growth contribution of interior finish NFPA 286—15 and critical radiant flux of floor covering NFPA 253—15 as listed	Interior wall/ceiling finish or floor covering charged to these tests	NFPA 286—15 is concatenated with the NFPA 285 row in the attached table	Conditional	Name NFPA 286—15 and NFPA 253—15 on finish specifications where those tests are charged; do not import flux or growth figures	Verified
UL 1784—01	Door-assembly air-leakage test	Air leakage tests of door assemblies as listed: UL 1784—01	Smoke or elevator-lobby door air-leakage testing	None stated	Conditional	Name UL 1784—01 on smoke and elevator-lobby door specifications; do not import leakage rates	Verified

8. Structure and envelope companions

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
SBC 304	Structural-concrete companion	Building Code Requirements for Structural Concrete as listed: SBC 304	Structural concrete	The attached SBC 304 table is page-split and cell-merged; designation and title remain readable. No edition year is printed in that cell	External verification	Name SBC 304 on structural-concrete specifications; do not import stresses, cover or detailing from SBC 304	Verified
SBC 305	Structural-masonry companion	Building Code Requirements for Masonry Structures as listed: SBC 305	Structural masonry	Companion masonry listings include TMS 402—2013 and TMS 602—2013	Conditional	Name SBC 305 if masonry is used structurally; do not import masonry stresses into this matrix	Verified
AISC 341—10	Structural-steel seismic edition	Seismic Provisions for Structural Steel Buildings as listed: AISC 341—10	Structural steel seismic design	None stated	Conditional	Name AISC 341—10 if structural steel is used; do not import seismic connection criteria	Verified
AA ADM1—2015; ASM 35—00	Structural-aluminum editions	Aluminum structures ADM1—2015 and aluminum sheet-metal work ASM 35—00 as listed	Structural aluminum	None stated	Conditional	Name ADM1—2015 and ASM 35—00 if structural aluminum is used; do not import allowable stresses or thicknesses	Verified
AISI S100—12	Cold-formed steel specification	North American Specification for the Design of Cold-formed Steel Structural Members, 2012 , as listed: AISI S100—12	Cold-formed steel structural members	Companion AISI framing standards are separately listed. AISI S230 is a one- and two-family method and is omitted	Conditional	Name AISI S100—12 if cold-formed steel is used; do not import member capacities. Do not apply AISI S230 to this tower	Verified
AAMA/WDMA/CSA 101/I.S.2/A440—11	Fenestration performance edition	Fenestration specifications for windows, doors and skylights as listed: AAMA/WDMA/CSA 101/I.S.2/A440—11	Windows, doors and skylights charged to this standard	Also listed under CSA and WDMA with the same designation	External verification	Name AAMA/WDMA/CSA 101/I.S.2/A440—11 on the fenestration specification; do not import performance grades from A440	Verified
ASTM E330—02; ASTM E331—00 (2009)	Curtain-wall pressure and water tests	Structural performance ASTM E330—02 and water penetration ASTM E331—00 (2009) of exterior windows, curtain walls and doors as listed	Exterior windows, curtain walls, skylights and doors	None stated	External verification	Name ASTM E330—02 and ASTM E331—00 (2009) on curtain-wall specifications; do not import pressure or leakage figures	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
FM 4474 (2011); UL 580—2006; UL 1897—12	Roof-assembly uplift editions	Roof-assembly wind-uplift tests as listed: FM 4474 (2011) , UL 580—2006 and UL 1897—12	Roof assemblies charged to these test methods	Edge systems ANSI/SPRI/FM 4435-ES-1—11 and ballasted single-ply ANSI/SPRI RP-4—13 are separately listed after a page-split SPRI header	Conditional	Name the listed FM and UL roof-uplift editions on the roof specification; do not import uplift values. Confirm SPRI rows against the printed code	Verified
Chapter 35 ASTM/UL tables	Listed material and test standards	ASTM, UL and related material, roofing, gypsum, fastener and test standards are listed by designation, effective date and citing section	Specified products that the charging chapters send to Chapter 35	Do not use title fragments such as 1 to 3 mm, 750°C, 345 MPa or 2,700 KN m/m³ as Chapter 35 design limits	External verification	Specify the Chapter 35 designation and year for each product; do not import test criteria or title-embedded quantities	Verified

9. Project-use controls

1. Use **Verified** rows for initial scoping after the row trigger and branch are confirmed.
2. Do not use the **Verify source ASTM F2006** year token as a design-release edition until the printed code is confirmed.
3. Do not import NFPA, ASME, ICC A117.1, ASTM, UL, SBC 304/305, TMS, AISI or AA criteria into drawings from this matrix.
4. Do not treat Chapter 35 as selecting **NFPA 13—13** versus **NFPA 13R—13** for Group R-2 high-rise.
5. Do not substitute a later A117.1, ASME A17.1 or NFPA 13 edition from memory.
6. Do not apply omitted not-typical listings (including **NFPA 13D—13**, **AISI S230**, **ICC 300/400/500/600**, aircraft hangars, fireworks, coal mines and one- and two-family wood manuals) unless the gap register opens that use.
7. Record locked companion editions and SASO/SBCNC adoption in the project Golden Thread; this matrix is not evidence of SCD NOC, SBPS approval or stamped compliance.

10. Coverage summary

Internal inventory of the attached Chapter 35 extract (numbered code, exceptions, tables, footnotes; commentary excluded). Row-level records are not published.

- **Inventory scope:** numbered code, exceptions, tables, footnotes (commentary excluded). Chapter 35 has no numbered 3501-style sections. Agency tables were read in full. General Comments and Purpose were excluded.
- **Total independently checkable numeric records:** 0
- **Verified:** 0
- **Verify source:** 0

Edition years, standard designation numbers and citing-section numbers were treated as identifiers, not independently checkable dimensional design rules. Title-embedded quantities were not inventoried as Chapter 35 limits.

Counts by top-level section

Top-level section	Records
Preamble (list identity, 102.4, SASO/SBCNC)	0
Agency referenced-standard tables	0

Coverage cross-check against SBC 201 Chapter 35 Referenced Standards (2024)_CS.md was topics-only: Chapter 35 is an edition list applied through **Section 102.4**, with SASO/SBCNC adoption for enforcement; architect-facing companions include **ICC A117.1—09**, **ASME A17.1—13/CSA B44—2013**, and **NFPA 13—13 / 13R—13**. No CS.md value was copied into a matrix cell.

Unrelated occupancy listings (including **NFPA 13D—13**, **AISI S230**, **ICC 300/400/500/600**, **IWUIC**, aircraft hangars, coal mines, fireworks, dry cleaning, cellulose nitrate, agricultural dust, cotton bales, belt manlifts and automotive lifts) were read as identifiers and omitted from lead tables.

11. Unresolved-source register

Agency tables in the attached extract are flattened HTML with page-split continuations. Identifiers used in project-use rows were adopted only where designation, year and title remained unambiguous. No dimensional value was repaired from memory.

Affected table / clause	Why unverified	Control note
ASTM F2006	Attached token reads F2006—00 (2005) 10 (year concatenation)	Do not adopt that year token as a design-release edition; confirm the printed ASTM F2006 year in the code volume
NFPA 720 citing cells	Attached cite token 9151.6.1 is OCR-damaged	Designation NFPA 720—15 was used; confirm citing sections in Chapter 9, not from this cell
SBC 304 table (pages 2007–2008)	Page-split and cell-merge	Designation SBC 304 and title were used; citing-section list was not adopted as a value
NFPA 285 / 286 row	Concatenated in the attached table	Designations NFPA 285—12 and NFPA 286—15 remain readable; do not import truncated cite lists
SPRI table	Empty header then content on the following page	SPRI designations on the following page were not used as design-release values in the roof-uplift row
ISO, PTI, TPI headers	Empty table then content on the following page	Those listings are not in the project-use tables
UL / ULC / USC / WCLIB / WDMA / WRI last page	Mega-merged table	Project-use UL rows were taken from earlier, readable UL pages only
Title fragments C954, E136, A588, D1557	1 to 3 mm, 750°C, 345 MPa, 2,700 KN m/m³ sit in outbound standard titles	Not Chapter 35 design limits; do not place them on drawings from this matrix